ISP Router

```
enable
conf t

hostname ISP
no ip domain-lookup

int gi0/0
ip address 203.0.113.1 255.255.255.252
no shutdown
exit

#static routing

ip route 10.255.0.0 255.255.255.252 203.0.113.2
ip route 10.255.0.4 255.255.255.252 203.0.113.2
ip route 10.10.0.0 255.255.0.0 203.0.113.2

enable secret cisco
service password-encryption
banner moth # UNAUTHORIZED ACCESS PROHIBITED #

SSH-
ip domain-name cisco.com
username cisco privilege 15 secret cisco123
crypto key generate rsa modulus 2048
ip ssh version 2
line vty 0 4
login local
transport input ssh
exit

end
wr
```

EDGE ROUTER

```
enable
conf t

hostname EDGE
no ip domain-lookup

int gi0/0
ip address 203.0.113.2 255.255.255.252
no shutdown
exit

int gi0/1
```

```
ip address 10.255.0.1 255.255.255.252
no shutdown
exit
int gi0/2
ip address 10.255.0.5 255.255.255.252
no shutdown
exit

#OSPF

router ospf 10
router-id 3.3.3.3
network 10.255.0.0 0.0.0.3 area 0
network 10.255.0.4 0.0.0.3 area 0
exit

enable secret cisco
service password-encryption
banner motd #UNAUTHORIZED ACCESS PROHIBITED #

#ssh
ip domain-name cisco.com
username cisco privilege 15 secret cisco123
crypto key generate rsa modulus 2048
ip ssh version 2
line vty 0 4
login local
transport input ssh
exit

#NAT_ACL

access-list 1 permit 10.10.0.0 0.0.255.255
int gi0/0
ip nat outside
exit

int gi0/1
ip nat inside
exit

int gi0/2
ip nat inside
exit

#PAT

ip nat inside source list 1 interface gigabitEthernet0/0 overload

#NTP
```

```
ntp server 10.10.40.10

#SYSLOG

logging host 10.10.40.20
logging trap informational
logging buffered 16384 informational
service timestamps log datetime msec

end
wr
---------------------------------------------------

CORE1

enable
conf t

hostname CORE1
no ip domain-lookup

int gi0/0
no switchport
ip address 10.255.0.2 255.255.255.252
no shutdown
exit

#OSPF

router ospf 10
router-id 1.1.1.1
network 10.255.0.0 0.0.0.3 area 0
exit


enable secret cisco
service password-encryption
banner motd #UNAUTHORIZED ACCESS PROHIBITED #

#ETHERCHANNEL
int range gi0/1 - 2
no switchport
channel-group 1 mode active
no shutdown
exit

int port-channel 1
no switchport
ip address 10.255.0.9 255.255.255.252
```

```
no shutdown
exit

#VLAN

vlan 10
name HR
vlan 20
name SALES
vlan 30
name IT
vlan 40
name SERVER
vlan 50
name GUEST
vlan 60
name SOC
vlan 99
name MGMT

#SVI

ip routing

int vlan 10
ip address 10.10.10.2 255.255.255.0
no shutdown
exit

int vlan 20
ip address 10.10.20.2 255.255.255.0
no shutdown
exit

int vlan 30
ip address 10.10.30.2 255.255.255.0
no shutdown
exit

int vlan 40
ip address 10.10.40.2 255.255.255.0
no shutdown
exit

int vlan 50
ip address 10.10.50.2 255.255.255.0
no shutdown
exit

int vlan 60
```

```
ip address 10.10.60.2 255.255.255.0
no shutdown
exit

int vlan 99
ip address 10.10.99.2 255.255.255.0
no shutdown
exit

#HSRP

int vlan 10
standby 10 ip 10.10.10.1
standby 10 priority 110
standby 10 preempt
exit

int vlan 20
standby 20 ip 10.10.20.1
standby 20 priority 110
standby 20 preempt
exit

int vlan 30
standby 30 ip 10.10.30.1
standby 30 priority 110
standby 30 preempt
exit

int vlan 40
standby 40 ip 10.10.40.1
standby 40 priority 110
standby 40 preempt
exit

int vlan 50
standby 50 ip 10.10.50.1
standby 50 priority 110
standby 50 preempt
exit

int vlan 60
standby 60 ip 10.10.60.1
standby 60 priority 110
standby 60 preempt
exit

int vlan 99
standby 99 ip 10.10.99.1
standby 99 priority 110
```

```
standby 99 preempt
exit

#STP

spanning-tree mode rapid-pvst
spanning-tree vlan 10,20,30,40,50,60,99

#DHCP_HELPER

int vlan 10
ip helper-address 192.168.40.1
exit

int vlan 20
ip helper-address 192.168.40.1
exit

int vlan 30
ip helper-address 192.168.40.1
exit

int vlan 40
ip helper-address 192.168.40.1
exit

int vlan 50
ip helper-address 10.10.40.10
exit

int vlan 60
ip helper-address 10.10.40.10
exit

#VLAN_INTO_OSPF

router ospf 10
network 10.10.10.0 0.0.0.255 area 0
network 10.10.20.0 0.0.0.255 area 0
network 10.10.30.0 0.0.0.255 area 0
network 10.10.40.0 0.0.0.255 area 0
network 10.10.50.0 0.0.0.255 area 0
network 10.10.60.0 0.0.0.255 area 0
network 10.10.99.0 0.0.0.255 area 0

#GUEST_ACL

ip access-list extended GUEST
 10 permit udp any eq bootpc any eq bootps
 20 deny ip 10.10.50.0 0.0.0.255 10.10.10.0 0.0.0.255
```

```
 30 deny ip 10.10.50.0 0.0.0.255 10.10.20.0 0.0.0.255
 40 deny ip 10.10.50.0 0.0.0.255 10.10.30.0 0.0.0.255
 50 deny ip 10.10.50.0 0.0.0.255 10.10.40.0 0.0.0.255
 60 deny ip 10.10.50.0 0.0.0.255 10.10.60.0 0.0.0.255
 70 deny ip 10.10.50.0 0.0.0.255 10.10.99.0 0.0.0.255
 80 permit ip 10.10.50.0 0.0.0.255 any

int vlan 50
ip access-group GUEST in

#NTP

ntp server 10.10.40.10

#SYSLOG

logging host 10.10.40.20
logging trap informational
logging buffered 16384 informational
service timestamps log datetime msec

end
wr

-------------------------------------------------

CORE2

enable
conf t

hostname CORE2
no ip domain-lookup

int gi0/0
no switchport
ip address 10.255.0.6 255.255.255.252
no shutdown
exit

#OSPF
router ospf 10
router-id 2.2.2.2
network 10.255.0.4 0.0.0.3 area 0
exit

enable secret cisco
service password-encryption
banner motd #UNAUTHORIZED ACCESS PROHIBITED #
```

```
#ETHERCHANNEL

int range gi0/1 - 2
no switchport
channel-group 1 mode active
no shutdown
exit

int port-channel 1
no switchport
ip address 10.255.0.10 255.255.255.252
no shutdown
exit

#VLAN
vlan 10
name HR
vlan 20
name SALES
vlan 30
name IT
vlan 40
name SERVER
vlan 50
name GUEST
vlan 60
name SOC
vlan 99
name MGMT

#SVI

ip routing

int vlan 10
ip address 10.10.10.3 255.255.255.0
no shutdown
exit

int vlan 20
ip address 10.10.20.3 255.255.255.0
no shutdown
exit

int vlan 30
ip address 10.10.30.3 255.255.255.0
no shutdown
exit

int vlan 40
```

```
ip address 10.10.40.3 255.255.255.0
no shutdown
exit

int vlan 50
ip address 10.10.50.3 255.255.255.0
no shutdown
exit

int vlan 60
ip address 10.10.60.3 255.255.255.0
no shutdown
exit

int vlan 99
ip address 10.10.99.3 255.255.255.0
no shutdown
exit

#HSRP

int vlan 10
standby 10 ip 10.10.10.1
standby 10 priority 100
standby 10 preempt
exit

int vlan 20
standby 20 ip 10.10.20.1
standby 20 priority 100
standby 20 preempt
exit

int vlan 30
standby 30 ip 10.10.30.1
standby 30 priority 100
standby 30 preempt
exit

int vlan 40
standby 40 ip 10.10.40.1
standby 40 priority 100
standby 40 preempt
exit

int vlan 50
standby 50 ip 10.10.50.1
standby 50 priority 100
standby 50 preempt
exit
```

```
int vlan 60
standby 60 ip 10.10.60.1
standby 60 priority 100
standby 60 preempt
exit

int vlan 99
standby 99 ip 10.10.99.1
standby 99 priority 100
standby 99 preempt
exit

#STP

spanning-tree mode rapid-pvst
spanning-tree vlan 10,20,30,40,50,60,99

#DHCP_HELPER

int vlan 10
ip helper-address 10.10.40.10
exit

int vlan 20
ip helper-address 10.10.40.10
exit

int vlan 30
ip helper-address 10.10.40.10
exit

int vlan 40
ip helper-address 10.10.40.10
exit

int vlan 50
ip helper-address 10.10.40.10
exit

int vlan 60
ip helper-address 10.10.40.10
exit

#VLAN_INTO_OSPF

router ospf 10
network 10.10.10.0 0.0.0.255 area 0
network 10.10.20.0 0.0.0.255 area 0
network 10.10.30.0 0.0.0.255 area 0
```

```
network 10.10.40.0 0.0.0.255 area 0
network 10.10.50.0 0.0.0.255 area 0
network 10.10.60.0 0.0.0.255 area 0
network 10.10.99.0 0.0.0.255 area 0

#GUEST_ACL

ip access-list extended GUEST
 10 permit udp any eq bootpc any eq bootps
 20 deny ip 10.10.50.0 0.0.0.255 10.10.10.0 0.0.0.255
 30 deny ip 10.10.50.0 0.0.0.255 10.10.20.0 0.0.0.255
 40 deny ip 10.10.50.0 0.0.0.255 10.10.30.0 0.0.0.255
 50 deny ip 10.10.50.0 0.0.0.255 10.10.40.0 0.0.0.255
 60 deny ip 10.10.50.0 0.0.0.255 10.10.60.0 0.0.0.255
 70 deny ip 10.10.50.0 0.0.0.255 10.10.99.0 0.0.0.255
 80 permit ip 10.10.50.0 0.0.0.255 any

int vlan 50
ip access-group GUEST in
exit

#NTP
ntp server 10.10.40.10

#SYSLOG

logging host 10.10.40.20
logging trap informational
logging buffered 16384 informational
service timestamps log datetime msec

end
wr

--------------------------------------------------

SW1

enable
conf t

hostname SW1
no ip domain-lookup

enable secret cisco
service password-encryption
banner motd #UNAUTHORIZED ACCESS PROHIBITED #

#VLAN
```

```
vlan 10
name HR
vlan 20
name SALES
vlan 30
name IT
vlan 40
name SERVER
vlan 50
name GUEST
vlan 60
name SOC
vlan 99
name MGMT

#SW1 uplink

int range gi0/0 - 1
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk allowed vlan 10,20,30,40,50,60,99
no shutdown
exit

#HR_PC

int gi0/2
switchport mode access
switchport access vlan 10
spanning-tree portfast
no shutdown
exit

#SALES

int gi0/3
switchport mode access
switchport access vlan 20
spanning-tree portfast
no shutdown
exit

#SYSLOG

logging host 10.10.40.10
logging trap warning
service timestamps log datetime msec
logging on

#DHCP snooping
```

```
ip dhcp snooping
ip dhcp snooping vlan 10,20

int gi0/0
ip dhcp snooping trust
exit

int gi0/1
ip dhcp snooping trust
exit

int range gi0/2 - 3
ip dhcp snooping limit rate 10
exit

#PortFast+BPDU

int range gi0/2 - 3
spanning-tree portfast
spanning-tree bpduguard enable
exit

#STORM CONTROL

int range gi0/2 - 3
storm-control broadcast level 5.00
storm-control multicast level 5.00
storm-control unicast level 5.00
exit

#Port Security

int gi0/2
switchport mode access
switchport access vlan 10
switchport port-security
switchport port-security maximum 1
switchport port-security mac-address sticky
switchport port-security violation restrict

int gi0/3
switchport mode access
switchport access vlan 20
switchport port-security
switchport port-security maximum 1
switchport port-security mac-address sticky
switchport port-security violation restrict

#DAI
```

```
ip arp inspection vlan 10,20
int range gi0/0 - 1
ip arp inspection trust
exit

end
wr

-----------------------------------------------------

sw2

enable
conf t

hostname sw2
no ip domain-lookup

enable secret cisco
service password-encryption
banner motd #UNAUTHORIZED ACCESS PROHIBITED #

#VLAN

vlan 10
name HR
vlan 20
name SALES
vlan 30
name IT
vlan 40
name SERVER
vlan 50
name GUEST
vlan 60
name SOC
vlan 99
name MGMT

#cong_sw2

int range gi0/0 - 1
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk allowed vlan 10,20,30,40,50,60,99
no shutdown
exit

#IT_PC
```

```
int gi0/2
switchport mode access
switchport access vlan 30
spanning-tree portfast
no shutdown
exit

#SYSLOG

logging host 10.10.40.10
logging trap warning
service timestamps log datetime msec
logging on

#DHCP snooping

ip dhcp snooping
ip dhcp snooping vlan 30

int gi0/0
ip dhcp snooping trust
exit

int gi0/1
ip dhcp snooping trust
exit

int range gi0/2
ip dhcp snooping limit rate 10
exit
no ip dhcp snooping information option


#PortFast+BPDU

int gi0/2
spanning-tree portfast
spanning-tree bpduguard enable
exit

#STORM CONTROL

int gi0/2
storm-control broadcast level 5.00
storm-control multicast level 5.00
storm-control unicast level 5.00
exit

#Port Security
```

```
int gi0/2
switchport mode access
switchport access vlan 30
switchport port-security
switchport port-security maximum 1
switchport port-security mac-address sticky
switchport port-security violation restrict

#DAI

ip arp inspection vlan 30
int range gi0/0 - 1
ip arp inspection trust
exit


end
wr

-----------------------------------------------------

sw3

enable
conf t

hostname sw3
no ip domain-lookup

enable secret cisco
service password-encryption
banner motd #UNAUTHORIZED ACCESS PROHIBITED #

#VLAN

vlan 10
name HR
vlan 20
name SALES
vlan 30
name IT
vlan 40
name SERVER
vlan 50
name GUEST
vlan 60
name SOC
vlan 99
name MGMT
```

```
#cong_sw2

int range gi0/0 - 1
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk allowed vlan 10,20,30,40,50,60,99
no shutdown
exit

#SOC_PC

int gi0/3
switchport mode access
switchport access vlan 60
spanning-tree portfast
no shutdown
exit

#SERVER

int gi0/2
switchport mode access
switchport access vlan 40
no shutdown

#SYSLOG

logging host 10.10.40.10
logging trap warning
service timestamps log datetime msec
logging on

#DHCP snooping

ip dhcp snooping
ip dhcp snooping vlan 40,60

int gi0/0
ip dhcp snooping trust
exit

int gi0/1
ip dhcp snooping trust
exit

int gi0/2
ip dhcp snooping trust
exit
```

```
int gi0/3
ip dhcp snooping limit rate 10
exit

int gi1/0
ip dhcp snooping trust
exit

no ip dhcp snooping information option

#PortFast+BPDU

int gi0/3
spanning-tree portfast
spanning-tree bpduguard enable
exit

#STORM CONTROL

int gi0/3
storm-control broadcast level 5.00
storm-control multicast level 5.00
storm-control unicast level 5.00
exit

#Port Security

int gi0/3
switchport mode access
switchport access vlan 60
switchport port-security
switchport port-security maximum 1
switchport port-security mac-address sticky
switchport port-security violation restrict

#DAI

ip arp inspection vlan 40,60
int range gi0/0 - 2
ip arp inspection trust

exit


end
wr

-----------------------------------------------------

sw4
```

```
enable
conf t

hostname sw4
no ip domain-lookup

enable secret cisco
service password-encryption
banner motd #UNAUTHORIZED ACCESS PROHIBITED #

#VLAN

vlan 10
name HR
vlan 20
name SALES
vlan 30
name IT
vlan 40
name SERVER
vlan 50
name GUEST
vlan 60
name SOC
vlan 99
name MGMT

#conf_sw4

int range gi0/0 - 1
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk allowed vlan 10,20,30,40,50,60,99
no shutdown
exit

#guest_pc

int gi0/2
switchport mode access
switchport access vlan 50
spanning-tree portfast
no shutdown
exit

#SYSLOG

logging host 10.10.40.10
logging trap warning
```

```
service timestamps log datetime msec
logging on

#DHCP snooping

ip dhcp snooping
ip dhcp snooping vlan 50

int gi0/0
ip dhcp snooping trust
exit

int gi0/1
ip dhcp snooping trust
exit

int range gi0/2
ip dhcp snooping limit rate 10
exit

no ip dhcp snooping information option

#PortFast+BPDU

int gi0/2
spanning-tree portfast
spanning-tree bpduguard enable
exit

#STORM CONTROL

int gi0/2
storm-control broadcast level 5.00
storm-control multicast level 5.00
storm-control unicast level 5.00
exit

#Port Security

int gi0/2
switchport mode access
switchport access vlan 50
switchport port-security
switchport port-security maximum 1
switchport port-security mac-address sticky
switchport port-security violation restrict

#DAI

ip arp inspection vlan 50
```

```
int range gi0/0 - 1
ip arp inspection trust

end
wr
```

SERVER

```
enable
conf t

hostname SERVER
no ip domain-lookup

enable secret cisco
service password-encryption
banner motd #UNAUTHORIZED ACCESS PROHIBITED #

int gi0/0
ip address 10.10.40.10 255.255.255.0
no shutdown
exit

ip route 0.0.0.0 0.0.0.0 10.10.40.1

#DHCP_POOL

ip dhcp excluded-address 10.10.10.1 10.10.10.20
ip dhcp excluded-address 10.10.20.1 10.10.20.20
ip dhcp excluded-address 10.10.30.1 10.10.30.20
ip dhcp excluded-address 10.10.40.1 10.10.40.20
ip dhcp excluded-address 10.10.50.1 10.10.50.20
ip dhcp excluded-address 10.10.60.1 10.10.60.20

#HR_pool

ip dhcp pool HR
network 10.10.10.0 255.255.255.0
default-router 10.10.10.1
dns-server 8.8.8.8
exit

#SALES_POOL

ip dhcp pool SALES
network 10.10.20.0 255.255.255.0
default-router 10.10.20.1
dns-server 8.8.8.8
```

```
exit

#IT_POOL

ip dhcp pool IT
network 10.10.30.0 255.255.255.0
default-router 10.10.30.1
dns-server 8.8.8.8
exit

#SERVER_POOL

ip dhcp pool SERVER
network 10.10.40.0 255.255.255.0
default-router 10.10.40.1
dns-server 8.8.8.8
exit

#GUEST_POOL

ip dhcp pool GUEST
network 10.10.50.0 255.255.255.0
default-router 10.10.50.1
dns-server 8.8.8.8
exit

#SOC_POOL

ip dhcp pool SOC
network 10.10.60.0 255.255.255.0
default-router 10.10.60.1
dns-server 8.8.8.8
exit

#NTP

ntp master 3


end
wr
```